Biomimicry for Aerospace applications

Exploratory concepts

This proposal connects my doctoral research in parasitology and immunology with potential applications in space debris removal, robotic grasping, planetary mobility and extraterrestrial sample collection.

Introduction

During my PhD, I investigated interactions between parasitic nematodes, their endosymbiotic bacteria and their vertebrate hosts. This work gave me an opportunity to study biological structures and strategies that help organisms function in challenging environments. Among the challenges facing space exploration, space-debris removal and planetary exploration particularly caught my attention. This proposal outlines four concepts inspired by organisms and biological mechanisms I encountered during my research. They are exploratory ideas; each would require further literature review, feasibility assessment and engineering development before an application could be established.

1. Grasping inspired by nematodes

Filaria are parasitic nematodes with a cylindrical, non-segmented, cuticle-covered body. Due to the pressure of internal fluids, the morphological structure of filaria is uniform and always round. The liquid contained in their internal cavity acts as a hydrostatic skeleton on which muscles operate. For reproduction, males wrap their tails around females to attach to them. This grasping method is enhanced by a system of self-gripping fasteners called the area rugosa (see Figure 1). The area rugosa consists of a ventral band of small longitudinal crests that prevents the female from slipping away.

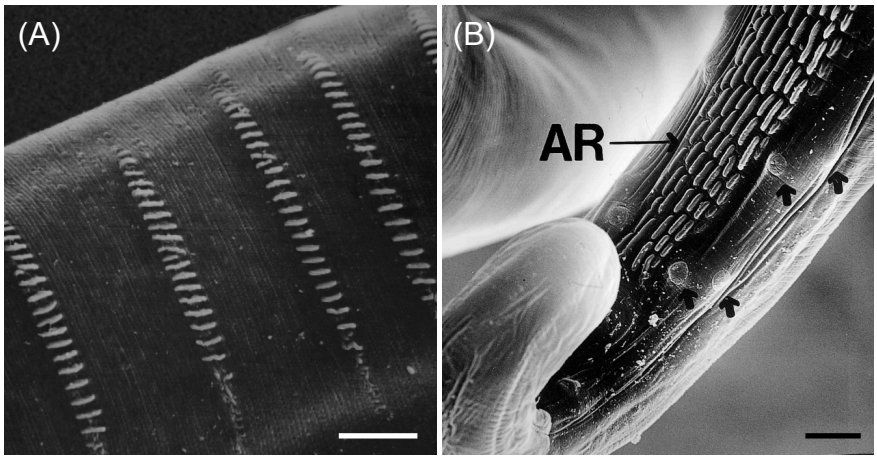


Figure 1: Male area rugosa of (A) *Litomosoides chagasfilhoi* (scale bar = 5 µm; from Muniz-Pereira et al., 2016) and (B) *Similascarophis maulensis* (scale bar = 10 µm; from Muñoz et al., 2004).

The organization of these crests may offer a solution to a common issue with harpoons, i.e. that the force required to penetrate the target's surface material is too high and pushes the target away (Banken and Oeffner, 2023). Among filaria, these self-gripping regions have diverse organizations for the same function (Figure 1, A and B). The area rugosa of several species could be studied by scanning electron microscopy to find a compromise that works in space conditions.

Liquid-filled fibers or capsules are already being studied for their self-repairing properties (Speck and Speck, 2019), and studying the mechanics of muscles acting on liquid-filled structures in worms could also lead to new perspectives for grasping devices.

2. Mimicking worm "dance" for planetary exploration

Filaria have longitudinal muscles divided into two dorsal and two ventral fields. The unequal contractions of the muscle cells in these four fields enable filaria to move by undulating rapidly. This so-called filarial "dance" allows them to move around in various tissues of their vertebrate host and prevents immune cells from sticking to and immobilizing them.

From the perspective of planetary exploration, this locomotion principle can be explored and may be useful for navigating diverse substrates. We can learn from worms by filming them on substrates we might encounter and using simulations. If muscle contractions can be reproduced and applied to a robot, this means of transport could prevent it from getting trapped.

3. Nets mimicking DNA traps

Several types of immune cells release DNA into the extracellular space, forming traps. Among them, extracellular traps produced by neutrophils are the most studied type of DNA trap. They are involved in the immobilization of bacteria and pathogens, allowing them to be cleared by other cells afterwards. Complex organisms pack their DNA into the limited space of their cellular nuclei by organizing it into chromatin, a complex of DNA and packaging proteins. Chromatin can condense DNA up to 10,000-fold compared to its free form. To release the net, histones untwist the DNA, and the nuclear and plasma membranes partially disintegrate. It has been shown that these nets can take diverse forms depending on the stimulus ("spiky" or "cloudy," for example; Daniel et al., 2019). Figure 2 shows a "spiky" neutrophil extracellular trap captured by Scanning Electron Microscopy.

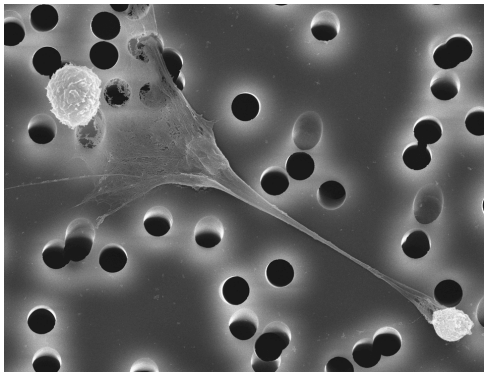


Figure 2: Image of a neutrophil extracellular trap upon stimulation by filaria, captured by Scanning Electron Microscopy. © Dr. Frédéric Fercoq, Team Parasites and Free Protists, Museum of Natural History.

The way DNA is condensed and untwisted by histones can be studied to understand the mechanisms shaping the net. In space, the trap could be released when approaching debris, and possibly several nets could be sent to collect larger debris. Because of its importance in gene expression, chromatin structure is widely studied. Studying interactions between packaging proteins and DNA requires the use of specific fluorescent markers and electron microscopy. The release of the net can be easily triggered in neutrophils in vitro to study the phenomenon.

4. Mite-inspired expandable matrix

Larval stages of filaria are transmitted to their hosts by blood-feeding insects like mites or mosquitoes during a blood meal. These insects ingest volumes of blood several times their pre-meal body weight. To accommodate this volume, the insect's membrane or cuticle undergoes plasticization. In most insects, the peritrophic matrix is the membrane that surrounds the food bolus (see Figure 3) and is formed of proteins and chitin microfibrils within a proteoglycan matrix.

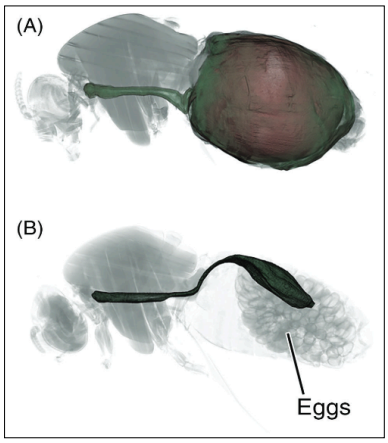


Figure 3: False-colored 3D images from CT imaging the peritrophic matrix of blackflies (A) 40 minutes or (B) 7.2 days after completing a blood meal (from Hall et al., 2021).

In space, this nature-inspired technology could be used for sample collection on planets or other extraterrestrial bodies. Peritrophic matrix fibrils can appear in diverse configurations, such as the orthogonal arrangements in *Forficula auricularia* or the hexagonal arrangements in *Oryctes nasicornis* (Lehane, 1997; see Figure 4, A and B respectively). By negatively staining individual peritrophic matrix microfibrils across different insect species, their diameter, length, and organization can be determined to establish the functional significance of these arrangements. Confocal imaging and micro-CT imaging can also be used to visualize the matrix's organization.

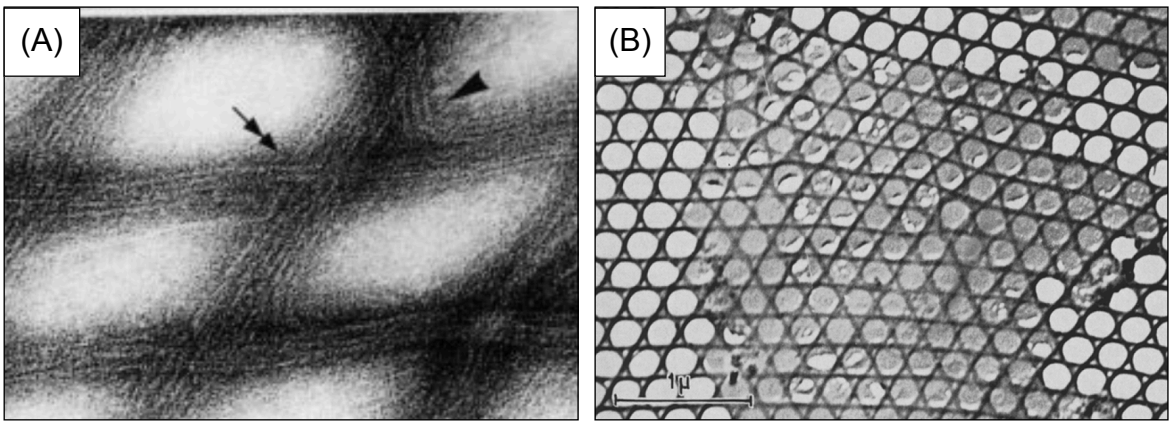


Figure 4: (A) The orthogonal arrangement of microfibrils in the peritrophic matrix of *Forficula auricularia* (x270 000, from Lehane, 1997). (B) Honeycomb organization of fibrils in the peritrophic membrane of *Oryctes nasicornis* (from Peters et al., 1979).

References

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